Abdul Rehman

Multivariable Calculus

BS-CS-Sem-II

Domain and Range

Topic:-
(James Stewart seventh edition)
Exercise 14.1, Book page 913

Question no 9:-

Let $g(x,y) = \cos(x+2y)$

a) Evaluate $g(2,-1)$

$g(2,-1) = \cos(2+2(-1))$

$g(2,-1) = \cos(2-2) = \cos(0)$

$g(2,-1) = 1$

b) Domain of g

The cosine function is defined for all real numbers. So the domain of function g is $\mathbb{R}^2$.

c) Range of g

The cosine function has a range of $[-1,1]$.

For the expression $x+2y$, there are no restrictions on its value since both x and y can take any real value.

Therefore, the range of $\cos(x+2y)$ is also $[-1,1]$.

Question no 10:-

$$F(x,y) = 1 + \sqrt{4-y^2}$$

a) Evaluate $F(3,1)$

$$F(3,1) = 1 + \sqrt{4-1^2}$$

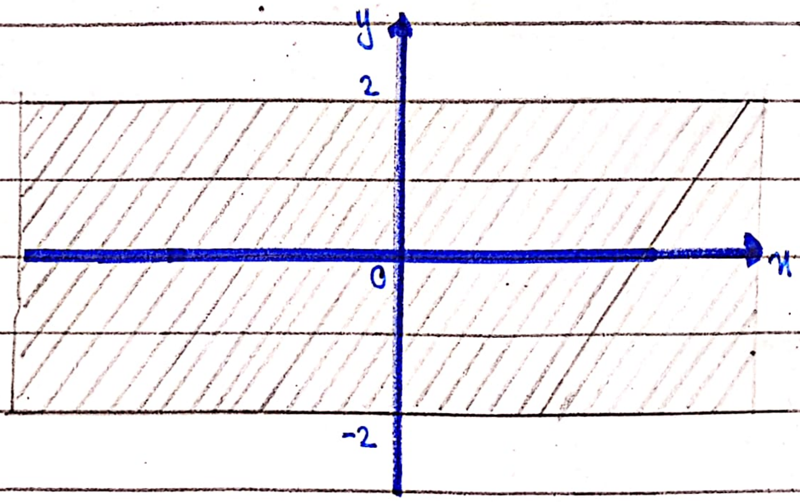
$$F(3,1) = 1 + \sqrt{3}$$

b) Find and sketch domain:-

$\sqrt{4-y^2}$ is defined only when $4-y^2 \geq 0$, or $y^2 \leq 4$.

So the domain of F is

$$\{(x,y) \mid -2 \leq y \leq 2\}$$



c) Range of function:-

We know, the range of $\sqrt{4-y^2}$ is $[0,2]$, by adding 1 doesn't change the range, it just shifts it up by 1. So, the range of $F(x,y) = 1 + \sqrt{4-y^2}$ is $[1,3]$.

Question no 11:-

$f(x,y,z) = \sqrt{x} + \sqrt{y} + \sqrt{z} + \ln(4 - x^2 - y^2 - z^2)$

a) **Evaluate** $f(1,1,1)$

$f(1,1,1) = \sqrt{1} + \sqrt{1} + \sqrt{1} + \ln(4-1-1-1)$

$f(1,1,1) = 3 + \ln(1)$

$f(1,1,1) = 3$

b) **Domain of f**

$\{(x,y,z) \mid x^2 + y^2 + z^2 < 4, x \geq 0, y \geq 0, z \geq 0\}$

This describes a region in 3D space where all there variables are non-negative and the sum of their squares is less than 4.

Geometrically, this corresponds to the interior of a sphere which has radius 2 centered at the origin.

Question no 12:-

Let $g(x,y,z) = x^3 y^2 z \sqrt{10 - x - y - z}$

a) **Evaluate** $g(1,2,3)$

$g(1,2,3) = (1)^3 (2)^2 (3) \sqrt{10-1-2-3}$

$g(1,2,3) = (4)(3)\sqrt{10-6}$

$g(1,2,3) = (4)(3)\sqrt{4}$

$g(1,2,3) = (12)(2)$

$g(1,2,3) = 24$

b) **Domain of function:-**

$\{(x, y, z) \mid x + y + z \leq 10\}$

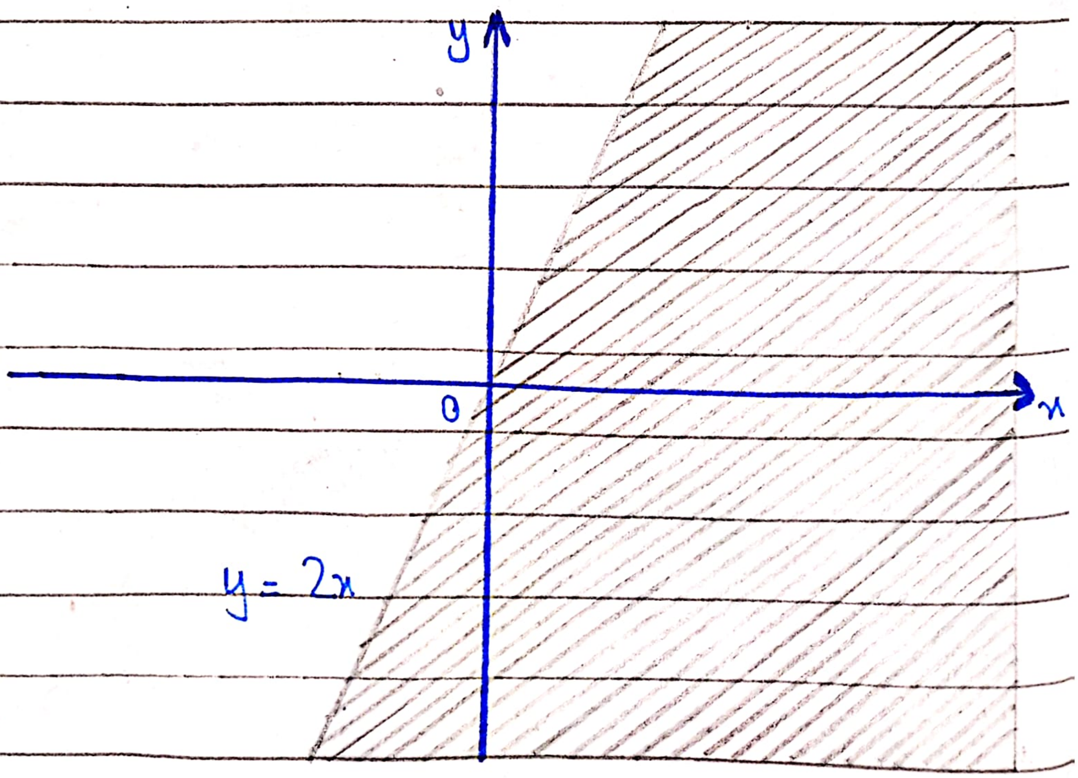
The domain of the function lies on or below the plane. This means that any point in 3D-space whose coordinates satisfy the inequality $x + y + z \leq 10$ belongs to the domain.

Find and sketch the domain of the function. (Question 13 to 22).

13. $f(x, y) = \sqrt{2x - y}$

$\sqrt{2x - y}$ is defined only when $2x - y \geq 0$ or $y \leq 2x$.

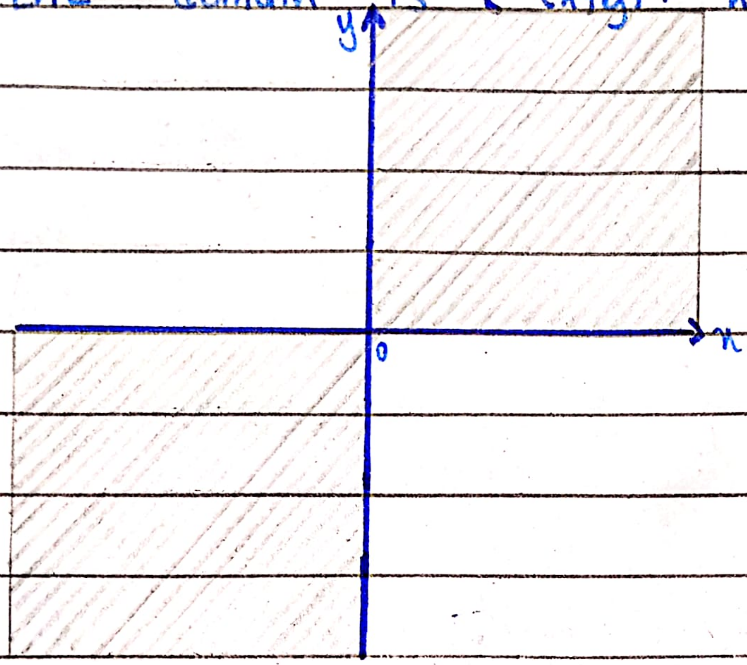
So, the domain of f is $\{(x, y) \mid y \leq 2x\}$



14. $f(x,y) = \sqrt{xy}$

$\sqrt{xy}$ is defined only when $xy \geq 0$.

So the domain is $\{(x,y) \mid xy \geq 0\}$



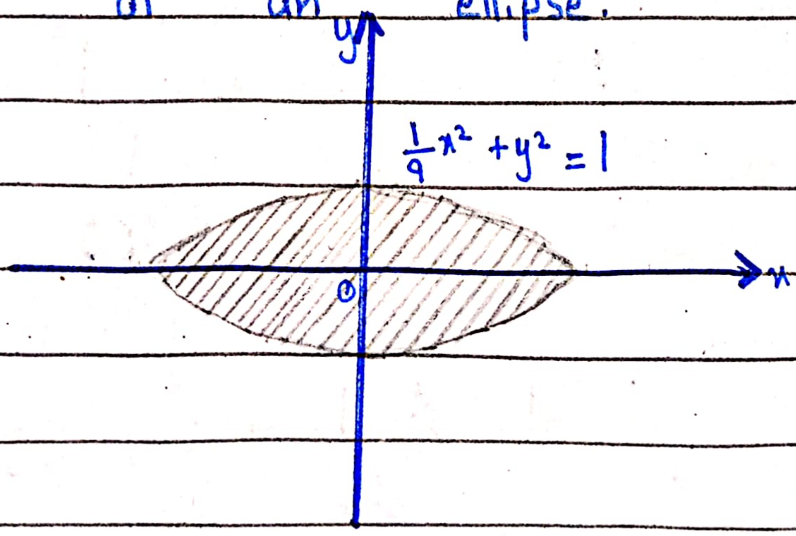
15. $f(x,y) = \ln(9 - x^2 - 9y^2)$

$\ln(9 - x^2 - 9y^2)$ is defined only when

$9 - x^2 - 9y^2 > 0$, or $\frac{x^2}{9} + y^2 < 1$

So the domain of f is

$\{(x,y) \mid \frac{x^2}{9} + y^2 < 1\}$, the interior of an ellipse.



16. $f(x,y) = \sqrt{x^2 - y^2}$

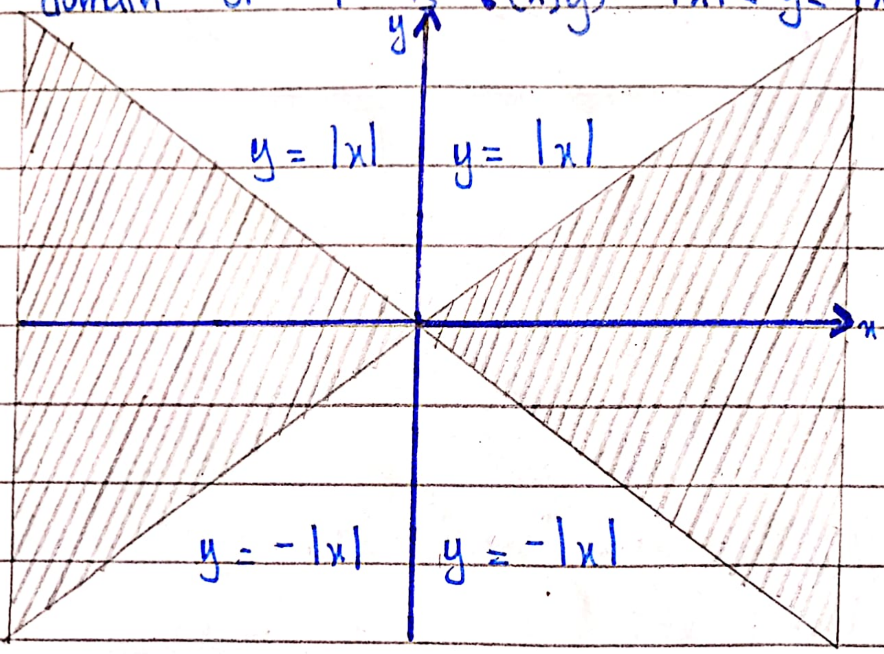
$\sqrt{x^2 - y^2}$ is defined only when $x^2 - y^2 \geq 0$

or $y^2 \leq x^2$

Which implies;

$$|y| \leq |x| \Leftrightarrow -|x| \leq y \leq |x|.$$

The domain of f is $\{(x,y) \mid -|x| \leq y \leq |x|\}$



17. $f(x,y) = \sqrt{1-x^2} - \sqrt{1-y^2}$

$\sqrt{1-x^2}$ is defined only when $1-x^2$ is greater or equal to zero, $1-x^2 \geq 0$,

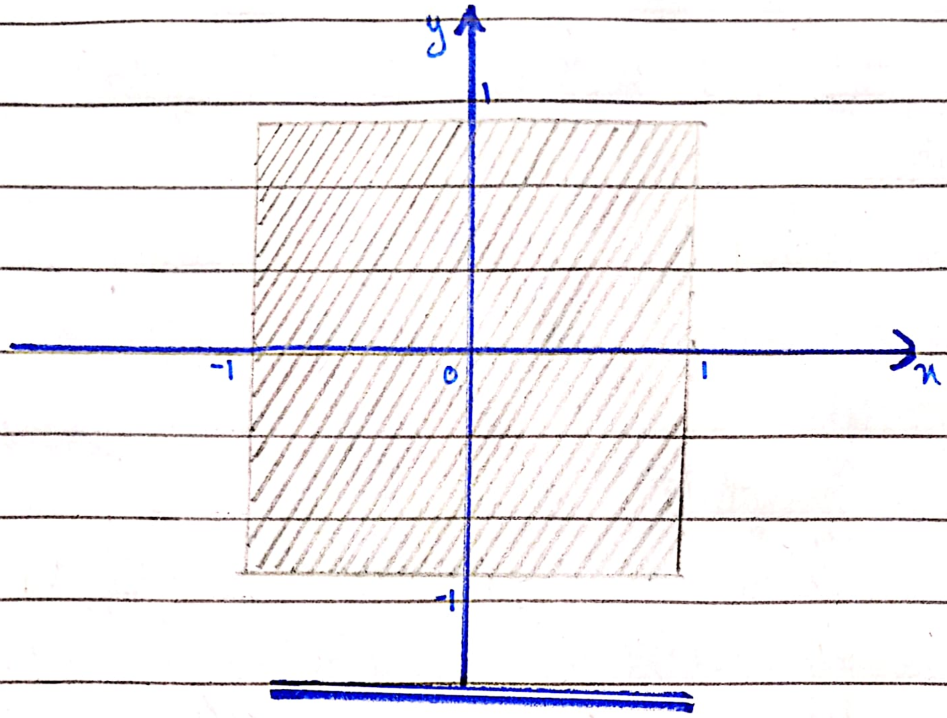
$x^2 \leq 1 \Leftrightarrow -1 \leq x \leq 1$, and

$\sqrt{1-y^2}$ is defined only when

$1-y^2 \geq 0$ or $y^2 \leq 1 \Leftrightarrow -1 \leq y \leq 1$.

Thus, the domain of f is

$\{(x,y) \mid -1 \leq x \leq 1, -1 \leq y \leq 1\}$

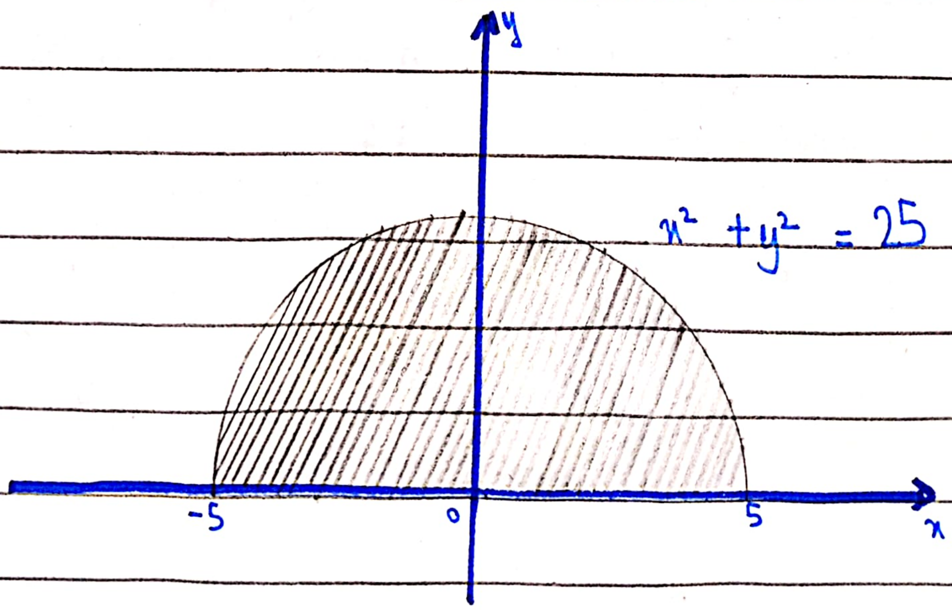


18. $f(x,y) = \sqrt{y} + \sqrt{25 - x^2 - y^2}$

$\sqrt{y} + \sqrt{25-x^2-y^2}$ is defined only when, $y \geq 0$ and $25 - x^2 - y^2 \geq 0$,

$25 \geq x^2 + y^2$.

So, domain is $\{(x,y) \mid y \geq 0,\ x^2+y^2 \leq 25\}$, a half disk of radius 5.

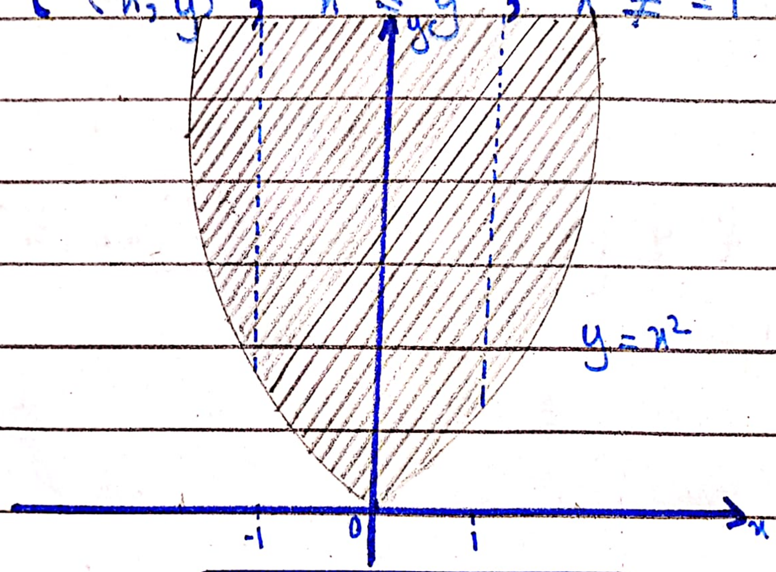


19. $f(x,y) = \dfrac{\sqrt{y-x^2}}{1-x^2}$

The function f is defined only when $y - x^2 \geq 0$, or $x^2 \leq y$ and $1 - x^2 \neq 0$, or $x^2 \neq 1$ or $x \neq \pm 1$.

Thus, the domain of f is

$$\{ (x,y) , x^2 \leq y , x \neq \pm 1 \}$$

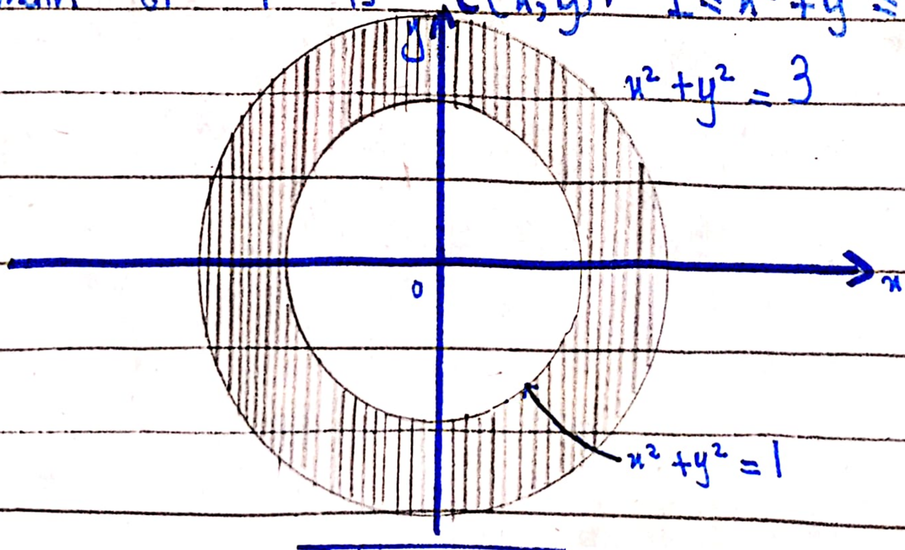


20. $f(x,y) = \sin^{-1}(x^2 + y^2 - 2)$

$\sin^{-1}(x^2+y^2-2)$ is defined when ~~x^2+y^2-2 is~~

$-1 \leq x^2+y^2-2 \leq 1 \iff 1 \leq x^2+y^2 \leq 3$.

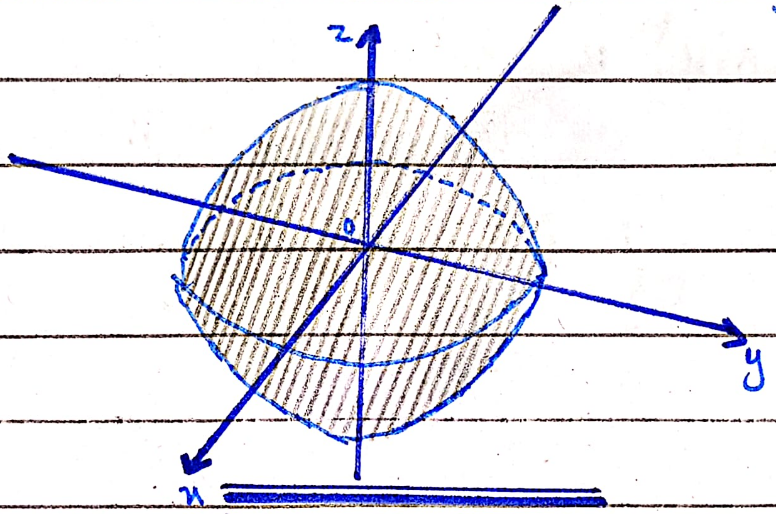
Domain of f is $\{(x,y) \mid 1 \leq x^2+y^2 \leq 3\}$.



21. $f(x,y,z) = \sqrt{1-x^2-y^2-z^2}$

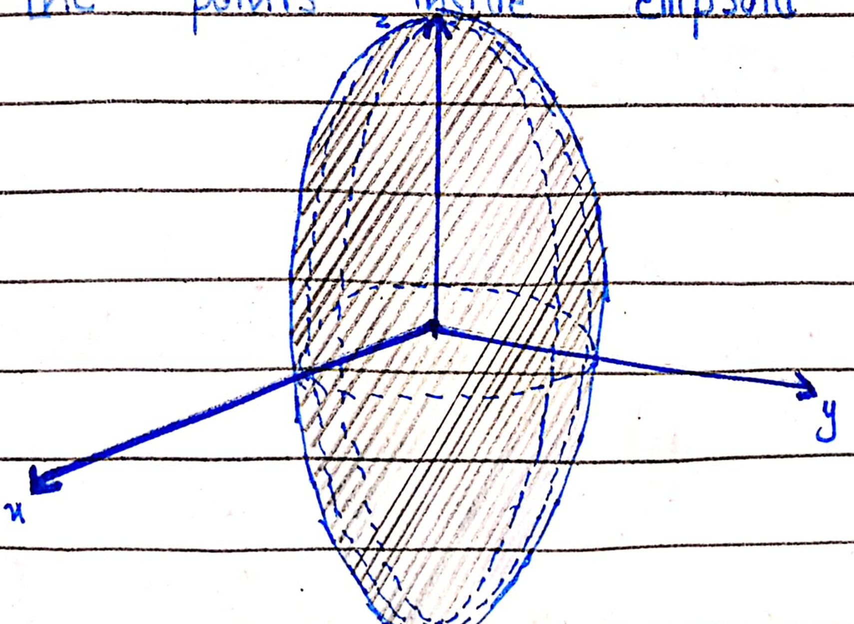
$\sqrt{1-x^2-y^2-z^2}$ is defined only when $1-x^2-y^2-z^2 \geq 0 \Leftrightarrow x^2+y^2+z^2 \leq 1$

So, the domain is $\{(x,y,z) \mid x^2+y^2+z^2 \leq 1\}$, The points inside or on the sphere of radius 1, center the origin.



22. $f(x,y,z) = \ln(16-4x^2-4y^2-z^2)$

f is defined only when $16-4x^2-4y^2-x^2>0$, or $\frac{x^2}{4} - \frac{y^2}{4} - \frac{x^2}{16} < 1$. Thus $D = \{(x,y,z) \mid \frac{x^2}{4} + \frac{y^2}{4} + \frac{z^2}{16} < 1\}$, that is, the points inside ellipsoid.



(Calculus by Swokowski)

Exercise 16.1, Book pg no 720:

Determine the domain of f and the value of f at the indicated points

1. $f(x,y) = 2x - y^2$

$f(x,y)$ is defined for all real numbers

$D = \{(x,y) \mid \mathbb{R}^2\}$

$f(-2,5)$

$f(-2,5) = 2(-2) - (5)^2$

$f(-2,5) = -4 - 25$

$f(-2,5) = -29$

$f(5,-2)$

$f(5,-2) = 2(5) - (-2)^2$

$f(5,-2) = 10 - 4$

$f(5,-2) = 6$

$f(0,-2)$

$f(0,-2) = 2(0) - (-2)^2$

$f(0,-2) = -4$

2. $f(x,y) = (y+2)/x$

$D = \{(x,y) \mid x \neq 0\}$

$f(3,1)$

$f(3,1) = (1+2)/3$

$f(3,1) = 1$

$f(3,1)$

$f(1,3) = (3+2)/1$

$f(1,3) = 5$

$f(2,0)$

$f(2,0) = (0+2)/2$

$f(2,0) = 1$

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3. $f(u,v) = uv/(u-2v)$

$D = \{ (u,v) \mid u \neq 2v \}$

$f(2,3)$

$f(2,3) = (2)(3)/(2-2(3))$

$f(2,3) = 6/(2-6)$

$f(2,3) = -\frac{3}{2}$

$f(-1,4)$

$f(-1,4) = (-1)(4)/(-1-2(4))$

$f(-1,4) = -4/(-1-8)$

$f(-1,4) = -4/-9 = 4/9$

$f(0,1)$

$f(0,1) = (0)(1)/(0-2(1))$

$f(0,1) = 0/(-2)$

$f(0,1) = 0$

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4. $f(r,s) = \sqrt{1-r} - e^{r/s}$

$D = \{ (r,s) \mid r \leq 1 , s \neq 0 \}$

$f(1,1)$

$f(1,1) = \sqrt{1-1} - e^{1/1}$

$f(1,1) = -e$

$f(0,4)$

$f(0,4) = \sqrt{1-0} - e^{0/4}$

$f(0,4) = 1-1$

$f(0,4) = 0$

$f(-3,3)$

$f(-3,3) = \sqrt{1-(-3)} - e^{\frac{-3}{3}}$

$f(-3,3) = \sqrt{1+3} - e^{-1}$

$f(-3,3) = \sqrt{4} - e^{-1}$

$f(-3,3) = 2 - e^{-1}$

5. $f(x,y,z) = \sqrt{25 - x^2 - y^2 - z^2}$

$D = \{ (x,y,z) \mid x^2 + y^2 + z^2 \leq 25 \}$

$f(1,-2,2)$

$f(1,-2,2) = \sqrt{25 - (1)^2 - (-2)^2 - (2)^2}$

$f(1,-2,2) = \sqrt{25 - 1 - 4 - 4}$

$f(1,-2,2) = \sqrt{25 - 9}$

$f(1,-2,2) = 4$

$f(-3,0,2)$

$f(-3,0,2) = \sqrt{25 - (-3)^2 - (0)^2 - (2^2)}$

$f(-3,0,2) = \sqrt{25-9-0-4}$

$f(-3,0,2) = \sqrt{25-13}$

$f(-3,0,2) = \sqrt{12}$

$f(-3,0,2) = \sqrt{(2)^2(3)}$

$f(-3,0,2) = 2\sqrt{3}$

6. $f(x,y,z) = (2 + \tan x + y\sin z)$

Since $\tan x$ is undefined at $x = (\pi/2) + \pi n$

$D = \{(x,y,z) \mid x \neq (\pi/2) + \pi n\}$

$f(\pi/4, 4, \pi/6)$

$f\left(\frac{\pi}{4}, 4, \frac{\pi}{6}\right) = 2 + \tan\frac{\pi}{4} + 4\sin\frac{\pi}{6}$

$f\left(\frac{\pi}{4}, 4, \frac{\pi}{6}\right) = 2 + 1 + 4\left(\frac{1}{2}\right)$

$= 5$

$f(0,0,0)$

$f(0,0,0) = 2 + \tan 0 + 0\sin 0$

$f(0,0,0) = 0$

7. $f(r,s,v,p) = rs^2\tan v + 4sv\ln p$

$D = \{(r,s,v,p) \mid v \neq \frac{\pi}{2} + n\pi,\ p > 0\}$

$f(3,-1,\pi/4,e)$

$f(3,-1,\pi/4,e) = (3)(-1)^2\tan\frac{\pi}{4} + 4(-1)\left(\frac{\pi}{4}\right)\ln e$

$f\left(3,-1,\frac{\pi}{4},e\right) = 3 - \pi$

8. $f(x,y,u,v,w) = w\ln(x-y) - xue^{v/w}$

~~f(x, y, u, v, w) =~~

$$D = \{(x,y,u,v,w) \mid y < x,\ w \neq 0\}$$

$f(2,1,3,4,-1)$

$$f(2,1,3,4,-1) = (-1)\ln(2-1) - (2)(3)e^{-4}$$

$$= -6e^{-4}$$

Find the following for ques 9-12

a. $\dfrac{f(x+h,y) - f(x,y)}{h}$ and

b. $\dfrac{f(x,y+h) - f(x,y)}{h}$ where $h \neq 0$.

9. $f(x,y) = x^2 + y^2$

a. $= \dfrac{(x+h)^2 + y^2 - x^2 - y^2}{h}$

$$= \frac{x^2 + h^2 + 2xh + y^2 - x^2 - y^2}{h}$$

$$= \frac{h^2 + 2xh}{h}$$

$$= h + 2x$$

b. $= \dfrac{x^2 + (y+h)^2 - x^2 - y^2}{h}$

$$= \frac{x^2 + y^2 + h^2 + 2yh - x^2 - y^2}{h}$$

$$= \frac{2yh + h^2}{h}$$

$$= 2y + h$$

10. $f(x,y) = y^2 + 3x$

a. $= \frac{y^2 + 3(x+h) - y^2 - 3x}{h}$

$$= \frac{y^2 + 3x + 3h - y^2 - 3x}{h}$$

$$= \frac{3h}{h} = 3$$

b. $= \frac{(y+h)^2 + 3x - y^2 - 3x}{h}$

$$= \frac{y^2 + h^2 + 2yh + 3x - y^2 - 3x}{h}$$

$$= \frac{2yh + h^2}{h}$$

$$= 2y + h$$

11. $f(x,y) = xy^2 + 3x$

a. $= \frac{(x+h)y^2 + 3(x+h) - xy^2 - 3x}{h}$

$$= \frac{xy^2 + y^2h - xy^2 + 3h}{h}$$

$$= \frac{y^2h + 3h}{h}$$

$$= y^2 + 3$$

b. $$= \frac{x(y+h)^2 + 3x - xy^2 - 3x}{h}$$

$$= \frac{x(y^2 + h^2 + 2yh) + 3x - xy^2 - 3x}{h}$$

$$= \frac{xy^2 + xh^2 + 2xyh + 3x - xy^2 - 3x}{h}$$

$$= \frac{h(xh + 2xy)}{h}$$

$$= xh + 2xy$$

12. $f(x, y) = xy^3 + 4x^2 - 2$

a. $$\frac{(x+h)y^3 + 4(x+h)^2 - 2 - xy^3 - 4x^2 + 2}{h}$$

$$= \frac{xy^3 + y^3h + 4x^2 + 4h^2 + 8xh - xy^3 - 4x^2}{h}$$

$$= \frac{y^3h + 4h^2 + 8xh}{h}$$

$$= y^3 + 8x + 4h$$

b. $$= \frac{x(y+h)^3 + 4x^2 - 2 - xy^3 - 4x^2 + 2}{h}$$

$$= \frac{x(y^3 + h^3 + 3y^2h + 3h^2y + y^2h + yh^2) - xy^3}{h}$$

$$= \frac{xy^3 + xh^3 + 3xy^2h + 3xyh^2 + xy^2h + xyh^2 - xy^3}{h}$$

$$= \frac{xh^3 + 3xy^2h + 3xyh^2}{h}$$

$$= 3xyh + 3xy^2 + xh^2$$

Partial Derivative

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Multivariable Calculus

BS-CS-Sem-II